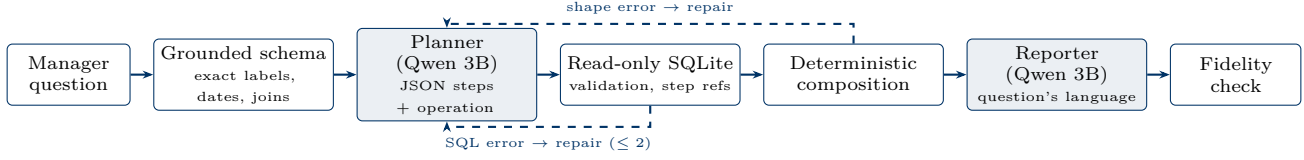


SQL4Business: Deliverable 2

Generative Artificial Intelligence (580694), Universidad de Concepción

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Code: github.com/VicenteSonez/Asistente-SQL-4-Business | Video: (video link pending)



Task, failure and model. Unchanged from Deliverable 1: answer punctual and combined business questions by executing the necessary SQL and reporting only figures faithful to the results. With direct prompting, all three candidates scored 0/5 on combined questions (invented years, translated labels, other SQL dialects, one query for a multi-value question). We commit to **Qwen/Qwen2.5-Coder-3B-Instruct** (revision 488639f, NF4 4-bit, bf16 compute, greedy decoding, Colab T4): the smallest candidate, tied for the best local baseline (46.7%); Qwen-7B scored 33.3% and Llama-8B also 46.7% with more parameters and gated access.

Intervention. Each part targets a diagnosed error. The grounded schema lists exact labels, the date range and join keys (translated labels, invented years). The planner decomposes the question into SQL steps and names one operation; code, not the model, computes growth, share, comparison and filter-then-rank (combined questions, mental arithmetic). SQLite runs read-only; validation or shape errors go back to the planner up to twice. A later step may read an earlier one (**FROM step₁**), expanded as a CTE. The reporter writes in the question's language, and the fidelity check accepts a figure only if it equals the verified answer or the evidence at the precision written, and checks the stated direction of a change.

Evaluation. Official set: the 15 Deliverable 1 questions (10 punctual, 5 combined), used while developing, so in-sample. Held-out set: 12 paraphrases (5/7) with the same gold, written during an audit and not used for design. Every system gets the same inputs and criteria: *sql* is the Deliverable 1 criterion (each gold result set produced exactly); *answer* checks the final answer, however it was computed; *report* and *e2e* (answer and report correct) apply to the solution. The baseline is the Deliverable 1 prompt verbatim; the alternative strategy adds only the grounded schema to it.

Results (table below). The solution answers 11/15 official and 10/12 held-out questions, against 8/15 and 5/12 for the baseline; on combined questions, 2/5 and 5/7 against 1/5 and 2/7. It answers and reports faithfully in 10/15 and 10/12. The grounded schema alone gives most punctual gains (9/10 official answers); decomposi-

tion adds the combined ones. Cost on the T4: 16.3 s per question against 6.2 s for the baseline.

Declared deviations. (1) The first version (17-09) chose the operation with keywords copied from the 5 combined questions and vetoed other plans: 12/15 in-sample but 0/7 on combined paraphrases. It was removed. (2) That version also used a new English baseline prompt (40%); the D1 prompt is restored. (3) The *answer* metric was added because the *sql* criterion rejects correct one-query answers (the baseline answers q14 correctly). (4) QLoRA fine-tuning, planned in D1, was not applied. (5) The first run of this version (v2) discarded plain-text reports as malformed JSON; v3 changes only the reporter's output format, and its SQL and answers are identical to v2 in 27/27 questions.

Failure case: q09. The planner filters `strftime('%Y-%m', sale_date) BETWEEN '2026-03-01' AND '2026-05-31'`. Comparing '2026-03' with a full date drops March and June, so the code computes 70.06% growth instead of 60.13%. The SQL is valid, the result is a single number and the report is faithful to it, so neither validation, repair nor the fidelity check can see the error: they check form and figures, not the meaning of a predicate. In q10 the model returns `strftime('%Q')` again after the error message, and in q09-p (held-out) it plans a single step, so the growth is never computed; its report invents 26%, 10% and 12%, and the check rejects it.

Limits. Official results are in-sample and both sets are small (27 questions, synthetic data). The fidelity check verifies figures, not their attribution: in q11 it accepts 488 reported as the total, and in q14-p a product id taken from a step. It is also conservative: it rejects "step 1" in q13. When the planner picks *direct* for a combined question, the answer depends on one query. Retries rarely fix a plan (strong model priors).

Reproduction. `notebooks/deliverable2_demo.ipynb` on a Colab T4 (*Run all*, about 25 min) clones the repository, runs the demo shown in the video and the evaluation, and writes `results/deliverable2_v3.json` and the table below; no uploads or tokens needed.

System	Metric	Official set (in-sample)			Held-out paraphrases	
		Punctual	Combined	All	Punctual	Combined
Direct prompt (D1 baseline)	sql	7/10	0/5	7/15	3/5	0/7
	answer	7/10	1/5	8/15	3/5	2/7
Direct prompt + grounded schema	sql	8/10	0/5	8/15	4/5	0/7
	answer	9/10	1/5	10/15	4/5	4/7
Structured solution	sql	9/10	2/5	11/15	5/5	3/7
	answer	9/10	2/5	11/15	5/5	5/7
	report	8/10	4/5	12/15	5/5	6/7
	e2e	8/10	2/5	10/15	5/5	5/7